§ 5.3 Diagonalization

Exmp 1 $D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$

$$D^2 = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix}$$

$$D^3 = \begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 5^3 & 0 \\ 0 & 3^3 \end{bmatrix}$$

In general, $D^k = \begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix}$ ☑

Def A: $n \times n$ matrix A is diagonalizable if A is similar to a diagonal matrix, that is if $A = PDP^{-1}$ for some invertible matrix P and some diagonal matrix D.

Characterization & Construction

★ Thm 5 (The diagonalization theorem)

A: $n \times n$ matrix A is diagonalizable iff A has n lin. ind. eigenvectors

In fact $A = PDP^{-1}$, w/ D a diagonal matrix,

iff the columns of P are n linearly independent eigenvectors of A.

Don't switch P & P^{-1} here

In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P.

Exmp Diagonalize, if possible,

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

That is, find an invertible matrix P and a diagonal matrix D s.t. $A = PDP^{-1}$.

Sol Step 1 eigenvalues

$$0 = \det(A-\lambda I) = \begin{vmatrix} 1-\lambda & 3 & 3 \\ -3 & -5-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{vmatrix}$$

$$= (1-\lambda)(-5-\lambda)(1-\lambda) - 27 - 27 - 9(-5-\lambda)$$

$$-(-9)(1-\lambda) - (-9)(1-\lambda)$$

$$= (-5-\lambda)(1-2\lambda+\lambda^2) - 54 + 45 + 9\lambda + 9 - 9\lambda + 9 - 9\lambda$$

$$= -5 + 10\lambda - 5\lambda^2 - \lambda + 2\lambda^2 - \lambda^3 + 9 - 9\lambda$$

$$= -\lambda^3 - 3\lambda^2 + 4$$

$$= -(\lambda-1)(\lambda^2+4\lambda+4) = -(\lambda-1)(\lambda+2)^2$$

eigenvalues: $\lambda = 1$, and $\lambda = -2$ (multiplicity 2)

Step 2 linearly independent eigenvectors

diagonalizable or not?

$\lambda = 1$ $\quad A - \lambda I = \begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 3 & 3 & 0 \end{bmatrix}$

$$\sim \begin{bmatrix} 3 & 3 & 0 \\ -3 & -6 & -3 \\ 0 & 3 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & -3 & -3 \\ 0 & 3 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

x_3 free variable. general sol: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ -x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ $\vec{v}_1$

eigenvector corr. to $\lambda = 1$

$\lambda = -2$ $\quad A - \lambda I = \begin{bmatrix} 3 & 3 & 3 \\ -3 & -3 & -3 \\ 3 & 3 & 3 \end{bmatrix}$

$$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

x_2, x_3 free variable

general solution: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_2 - x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

$\vec{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ $\vec{v}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

$\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ ~ linearly independent: eigenvectors corr. to $\lambda = -2$

$$\begin{vmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = 1 + 0 + 0 - (-1) - 1 - 0 = 1$$

Step 3 Construct P (invertible) from vectors in Step 2

$$P = [\vec{v}_1 \ \vec{v}_2 \ \vec{v}_3] = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Step 4 Construct D (diagonal) from the corr. eigenvalues.

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

(4)

Check $A = PDP^{-1}$

Exmp 4 Diagonalize $A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ if possible.

Sol Find eigenvalues:

$$0 = \det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 4 & 3 \\ -4 & -6-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{vmatrix} = -\lambda^3 - 3\lambda^2 + 4$$

$$= -(\lambda - 1)(\lambda + 2)^2$$

Find eigenvectors:

$\lambda = 1$

$$A - \lambda I = \begin{bmatrix} 1 & 4 & 3 \\ -4 & -7 & -3 \\ 3 & 3 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 4 & 3 \\ 0 & 9 & 9 \\ 0 & -9 & -9 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

↑ free var x_3

eigenvector: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ -x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ $x_3 \in \mathbb{R}$

Focus on eigenvalues w/ high multiplicities. This is where non-diagonal. may occur.

$\lambda = -2$ (multiplicity 2)

$$A - \lambda I = \begin{bmatrix} 4 & 4 & 3 \\ -4 & -4 & -3 \\ 3 & 3 & 3 \end{bmatrix} \sim \begin{bmatrix} 4 & 4 & 3 \\ 0 & 0 & 0 \\ 3 & 3 & 3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 \\ 4 & 4 & 3 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

↑ free var x_2

eigenvector:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_2 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \quad x_2 \in \mathbb{R}$$

There are only 2 linearly independent eigenvectors

NOT diagonalizable

Thm 6. An $n \times n$ matrix w/ n distinct eigenvalues is diagonalizable.

Rmk Not a necessary condition

Exmp 5. $A = \begin{bmatrix} 5 & -8 & 1 \\ 0 & 0 & 7 \\ 0 & 0 & -2 \end{bmatrix}$ ~ diagonalizable

Non-distinct eigenvalues

In the case of Thm 6, we can take the corresponding eigenvectors $\vec{v}_1, \cdots, \vec{v}_n$, and let

$$P = [\vec{v}_1, \cdots, \vec{v}_n]$$

$$P^{-1}AP = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

Non-distinct eigenvalues

Thm 7 A: $n \times n$ matrix whose distinct eigenvalues are $\lambda_1, \cdots, \lambda_p$

a) For $1 \le k \le p$, the dimension of the eigenspace (V_k) corresponding to λ_k $\le$ multiplicity of λ_k

b) A diagonalizable iff

$$\sum_{k=1}^{p} \dim V_k = n \quad \text{iff}$$

(i) $|A - \lambda I|$ factors completely into linear factors and

(ii) $\dim V_k$ = multiplicity of λ_k $\forall\, k = 1, \cdots, p$

c) If A is diagonalizable, and B_k is a basis for V_k then $\bigcup_{k=1}^{p} B_k$ forms an eigenvector basis for $\mathbb{R}^n$.

Exmp 6.